COUP DE GRACE
At the end of a high-speed chase, a Canadian lynx catches a snowshoe hare. Population cycles of both species are thought to be closely linked.

form the coat's water-repellent surface, while shorter, much denser, hairs—the underfur—trap a layer of air close to the body, keeping the animal warm. Northern species all grow an extra-thick coat after their late-summer molt; and some species, such as the stoat, use this molt to change color, developing a white coat that provides better camouflage for the winter.

Keeping warm is relatively easy for large mammals because their bodies contain a large store of heat. But for the smallest warm-blooded inhabitants of coniferous forest, winter conditions test their cold tolerance to its limits. Voles and other rodents can hide in burrows, but birds spend most of their lives in the open. For wrens and tits, which often weigh less than 3/8 oz (10 g), winter nights are a particularly dangerous time. With such minute bodies, their fuel reserves are tiny, and so they must make special provision if they are to stay alive until dawn when the search for food can resume. Some of them make the most of what body heat they have by huddling together in tree holes, but a few, such as the Siberian tit, bed down in the snow, using it as an insulating material.

RUNNING ON SNOW
Cold winters actually make it easier for the wolverine to catch food: its ability to run at high speed over frozen snow enables it to catch large animals, such as reindeer, which can outpace it during the summer months.

Population cycles

Since there are relatively few animal species in the northern coniferous forests, the lives of predators and prey are very closely linked. During mild years, strong tree growth can trigger a population explosion among small animals; as a result, the predators that feed on them begin to increase in number. These conditions never last for long, though: as the plant eaters begin to outstrip the food supply, their numbers start to fall again. And as the rate of the fall accelerates, the predators soon follow suit.

Despite the unpredictability of the northern climate, these ups and downs occur with surprising regularity. In North America, fur trappers' records dating back over a century provide some long-term evidence of population swings. For example, they show that the snowshoe hare population roughly follows a 10-year cycle, with 2 or 3 good years, followed by a lengthy slump. The Canadian lynx—one of the snowshoe hare's main predators—follows the same pattern, but with a one- or 2-year time-lag. Similar cycles involving lemmings and other small mammals take place in tundra (see p.67).

While there is little that they can do to prevent this boom-and-bust pattern from occurring, animals like the snowshoe hare are able to make a fairly fast recovery from a population slump by breeding quickly when conditions are favorable.

IRRUPTIONS

In the northern coniferous forests, food supplies are affected by the weather and the degree of competition. In winters that follow cool summers, the supply of seeds and berries can be thin, leaving seed- and fruit-eating birds with little to survive on. Rather than starve, these birds fly south in waves, called irruptions, which may involve traveling beyond their normal winter range by as much as 930 miles (1,500 km). Species that frequently irrupt include crossbills, waxwings, and tits, as well as nutcrackers and other seed-eating members of the crow family.

FLYING SOUTH
A flock of Bohemian waxwings pause during their southward flight. Like other irrupting species, waxwings usually spread south when their population reaches a high level. Shortage of winter food can trigger a mass exodus.